

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I: TWO PORT NETWORK THEORY

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6701 RF and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: RF behavior of Resistors, Capacitors and Inductors

Name of the Innovative Practice: Hardware Demonstration

Date & Time: 19.07.2019 & 20 minutes

ICT Tool Used:- Smart Class Room

Description of Hardware Demonstration:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

1. The students will be able to acquire better knowledge about RF behavior of Resistors, Capacitors and Inductors.
2. The students will be able to understand the low frequency circuit elements and high frequency circuit elements clearly.

Use of appropriate methods:

Justification for choosing the Hardware Demonstration:

The students use the Through Hole (TH) components in the laboratory experiments. They are not aware of RF behavior of Resistors, Capacitors and Inductors (**Surface Mounted Devices**). For that, I have chosen Hardware demonstration as an Innovative practice method. In that I have demonstrated **Raspberry Pi board SMD** components.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution	
RF behavior of Resistors, Capacitors and Inductors– Demonstration	
 Raspberry Pi board was shown to the students for demonstrating the RF behavior of Resistors, Capacitors and Inductors	 Students saw the RF resistors and capacitors in the Raspberry Pi board
 Students saw the RF resistors and capacitors in the Raspberry Pi board	 Students saw the RF resistors and capacitors in the Raspberry Pi board

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was assessed by asking questions related with RF basics.
- ❖ The assessment of effectiveness of the activity was felt when the students are related SMD components of Raspberry Pi board with RF passive devices.

Reflective Critique:

- **Benefits:**

The students confused about the low frequency passive devices with RF behaviour of Resistors, Capacitors and Inductors. After that, I have chosen Hardware Demonstration as an Innovative practice method for better understanding about RF behaviour of Resistors, Capacitors and Inductors. While demonstration, The Students can be seen Raspberry Pi board and they realize the RF behaviour of Resistors, Capacitors.

- **Challenges:**

Initially, I have planned the activity for 10 minutes but students are interested to see the Raspberry Pi board and they have compared Surface Mounted Devices (SMD) of Resistors, Capacitors with Low frequency passive devices which it takes another 10 minutes for completing the activity.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	1					2	1		

CO1: The students will be able to design and analyse multiport RF networks.

References:

1. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications", Pearson Education Inc., 2011
2. <https://www.raspberrypi.org/>

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UNIT II RF AMPLIFIERS AND MATCHING NETWORKS

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6701 RF and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Amplifier Power Relations

Name of the Innovative Practice: Mind map

Date & Time: 22.07.2019 & 15 Minutes

ICT Tool Used:- Smart Class Room

Description of Mind Map Technique:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

- ❖ The students will be able to differentiate the operating power gain, Available power gain, Transducer gain
- ❖ The students will be able to get better knowledge about the amplifier power relations
- ❖ The students will be able to identify the corresponding amplifier power relation equations for solving the problems

Use of appropriate methods:

Justification for choosing the Mind Map Activity:

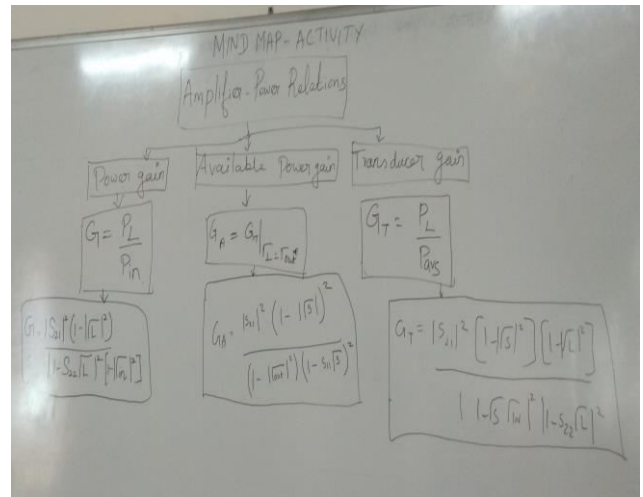
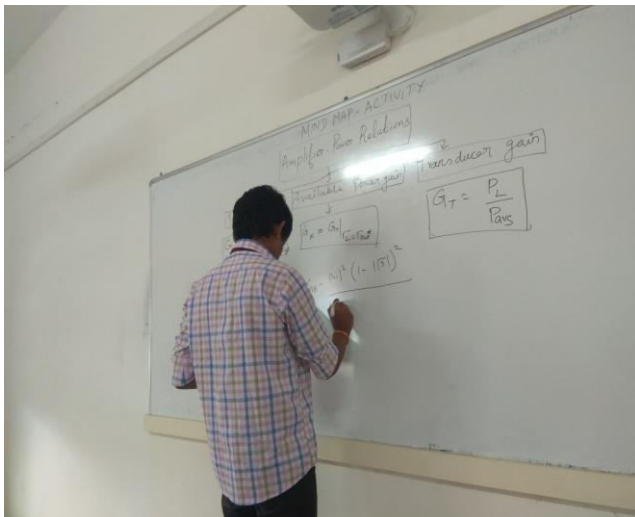
Mind map activity gives the graphical representation of Amplifier power relations. It differentiates the Power gain, Available power gain and Transducer Power gain clearly which will be helpful for understanding the concepts of amplifier power relations to the students.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution

Amplifier Power Relations:- Mind Map



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 70% of students answered correctly
- ❖ While conducting the activity, I understood that one point was missed with respect to Unilateral Transducer gain and finally I insisted the point they missed to tell
- ❖ The assessment of effectiveness of the activity was felt when told most of the points

Reflective Critique:

- **Benefits:**

The students confused about the different types of amplifier power gains (Operating Power gain, Available Power Gain and Transducer Power gain) which was derived in the class room. After that, I have chosen Mind Map as an Innovative practice method for better understanding about Amplifier power relations. The Students eagerly participated in that activity and then they acquired the concept of amplifier power relations.

- **Challenges:**

Initially, I have planned the activity for 10 minutes but students not aware of this activity. So, that I have explained the activity then some of the students hesitated for doing that. I encouraged the students which it takes another 5 minutes for completing the activity.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1					2	1		

CO2: The students will be able to formulate solutions for stability and impedance matching issues in microwave amplifiers

References:

1. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications", Pearson Education Inc., 2011
2. <https://www.mindmapping.com/mind-map.php>
3. <https://blog.iqmatrix.com/how-to-mind-map>

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UNIT III PASSIVE AND ACTIVE MICROWAVE DEVICES

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6701 RF and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Circulator, Isolator

Name of the Innovative Practice: Role Play

Date & Time: 26.08.2019 & 20 Minutes

ICT Tool Used:- Smart Class Room

Description of Role Play Technique:

Role-play is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Role play is an educational technique in which people spontaneously act-out problems of human relations and analyze the enactment with the help of other role players and observers.

Role playing is effective when the topic involves person to person communication or interactions. It can allow everyone to participate. The overall mood a training session can be improved by the excitement that role playing provide.

Goals (Learning Outcomes):

- ❖ The students will be able to acquire the better knowledge about circulator and isolator.
- ❖ The students will be able to differentiate the working principles of 3 port and 4 port circulators and 2 port isolators.

Use of appropriate methods:

Justification for choosing the Role Play Activity:

During the Role play activity, students know their roles about circulator and isolator working principles and operations. Some of the students acted as an isolator and some of them acted as circulator. It helps to differentiate the operation of circulator and isolator effectively among all other students.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution

Circulator, Isolator – Role Play



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly
- ❖ The assessment of effectiveness of the activity was felt when told most of the points

Reflective Critique:

- **Benefits:**

Most of the students confused about the operation and S-matrix properties of the circulator and isolator which was taught in the class room. After that, I felt Role play is the best activity for clearly explaining the concept of circulator and isolator. Some of the students interested for doing the activity. They have prepared and discussed about the operation of circulator and isolator effectively but it takes more time to complete it. At last, most of the students benefited and acquired better knowledge about it.

- **Challenges:**

I have planned the activity for 10 minutes but students asked more time for preparing their roles. It takes another 10 minutes to complete this activity.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	3	2	1					2	2		

CO3: The students will be able to explain the basic concept of various microwave devices

References:

1. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications", Pearson Education Inc., 2011
2. <https://www.teachingenglish.org.uk/article/role-play>
3. <https://www.mindtools.com/CommSkill/RolePlaying.htm>
4. <https://study.com/academy/lesson/what-is-role-play-definition-scenario-ideas-activities.html>

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UNIT IV MICROWAVE GENERATION

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6701 RF and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Traveling wave tube amplifier

Name of the Innovative Practice: Brainstorming Technique

Date & Time: 16.09.2019 & 15 Minutes

ICT Tool Used:- Smart Class Room

Description of Brainstorming Technique:

Brainstorming is a method for generating ideas to solve a design problem. It usually involves a group, under the direction of a facilitator. The strength of brainstorming is the potential participants have in drawing associations between their ideas in a free-thinking environment, thereby broadening the solution space.

The normal view of brainstorming is where a group of people sit in a room and shout out ideas as they occur to them. They are told to lose their inhibitions and that no ideas will be judged so that people are free to shout out any ideas at all without feeling uncomfortable. People should build on the ideas called out by other participants. The purpose of this is to gain as many ideas as possible for later analysis. Out of the many ideas suggested there will be some of great value. Because of the free-thinking environment, the session will help promote radical new ideas which break free from normal ways of thinking.

Goals (Learning Outcomes):

- ❖ The students will be able to acquire knowledge about operation of Traveling Wave Tube amplifier (TWT)
- ❖ The students will be able to understand the slow wave structure principle which is incorporated in the TWT amplifier and also they got clear idea about High negative potential of Repeller electrode

Use of appropriate methods:

Justification for choosing Brainstorming Activity:

Before the topic of Traveling wave tube amplifier, I have taught two cavity Klystron amplifier and Reflex Klystron oscillator. Therefore, students should know about microwave tubes. For that reason, I have chosen Brainstorming activity for this topic. Definitely, it helps to differentiate the operation of two cavity klystron amplifier and Travelling wave tube amplifier.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution Traveling wave tube amplifier - Brainstorming Technique



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ While conducting the activity, I understood that one point was missed with respect to method of excitation to the Travelling wave tube amplifier and finally I insisted the point they missed to tell
- ❖ The assessment of effectiveness of the activity was felt when told most of the points

Reflective Critique:

- **Benefits:**

Some of the students did silly mistakes while doing the activity. It takes more time to recall about microwave tubes which was taught in previous lectures. After that, students eagerly participated and brainstormed their views about travelling tube amplifier. However, some of them did mistakes but most of them explained correctly. At last, I concluded about the working principle and operation of travelling tube amplifier.

- **Challenges:**

I have planned the activity for 10 minutes but students struggled to recall about microwave tubes and then I gave another 5 minutes to recall it. Totally, it took 15 minutes to complete it.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	3	2	1					2	1		

CO4: The students will be able to elaborate the concepts involved in Microwave generation

References:

1. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications", Pearson Education Inc., 2011
2. <http://www.brainstorming.co.uk/tutorials/whatisbrainstorming.html>
3. <https://www.interaction-design.org/literature/topics/brainstorming>
4. <https://www.mindtools.com/brainstm.html>

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UNIT V MICROWAVE MEASUREMENTS

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6701 RF and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Measurement of Impedance, Frequency, Power

Name of the Innovative Practice: Demonstration

Date & Time: 27.09.2019 & 20 Minutes

ICT Tool Used:- Smart Class Room

Description of Demonstration:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

- ❖ The students will be able to acquire better knowledge about the measurement of Impedance, Frequency using Slotted Line Section
- ❖ The students will be able to operate the Slotted Line Section for measuring the indirect way of frequency

Use of appropriate methods:

Justification for choosing Brainstorming Activity:

In the Optical and Microwave laboratory, all the students did measurement of Impedance, Frequency, Power using slotted line section and VSWR meter. I want to correlate this experiment to theory course. For that, I want to demonstrate the slotted line section experiment for measuring Impedance, Frequency, and Power.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution Measurement of Impedance, Frequency, Power - Demonstration



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ While conducting the demonstration, students related the theoretical concepts of measurement of Impedance, Frequency, Power with laboratory experiments
- ❖ The assessment of effectiveness of the activity was felt when asked viva questions after demonstration

Reflective Critique:

- **Benefits:**

During the class, most of the students confused about measuring the impedance, frequency and power using Slotted line section and VSWR meter. After demonstration, they can be acquired better knowledge about measuring these parameters. Some of the students not interested during the demonstration then I counselled them separately and told the importance of hardware demonstration.

- **Challenges:**

I have planned the Slotted line section experiment demonstration nearly 20 minutes.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1					2	1		

CO5: The students will be able to measure and analyse Microwave signal and its parameters.

References:

1. Reinhold Ludwig and Gene Bogdanov, "RF Circuit Design: Theory and Applications", Pearson Education Inc., 2011
2. <https://www.youtube.com/watch?v=PSfBSWnzcDc>
3. <https://www.youtube.com/watch?v=Tn0qL-SY8PM>